

OA3103

Binary & Categorical Predictors

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ISLR: 3.3.1: Qualitative Predictors

TALR: 14.3: Categorical Predictors

Example

We build a linear model for IQ of NPS students using a qualitative predictor of whether the student is in the OR curriculum.

- ▶ Y : IQ, a quantitative variable
- ▶ X : Is the student in OR?

$$\begin{cases} 1 & \text{if yes} \\ 0 & \text{if no} \end{cases}$$

So

$$Y = \beta_0 + \beta_1 X + \epsilon.$$

and our estimate

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X$$

Interpretation

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- ▶ $\hat{Y}|\{\text{Student not in OR}\}$ is $\hat{Y}|X = 0$, which is $\hat{\beta}_0$.
- ▶ $\hat{Y}|\{\text{Student in OR}\}$ is $\hat{Y}|X = 1$, which is $\hat{\beta}_0 + \hat{\beta}_1$.

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- ▶ $\hat{Y}|\{\text{Student in OR}\}$ is $\hat{Y}|X = 1$, which is $\hat{\beta}_0 + \hat{\beta}_1$.

So $\hat{\beta}_1$ is an adjustment to the intercept when the student is in OR.

...we hope it is positive.

In General

To use membership in a category $\mathcal{C}$ as a predictor, we **code** it as an **indicator/dummy** variable X .

$$\begin{cases} X = 0 & \text{if observation not in } \mathcal{C} \\ X = 1 & \text{if observation in } \mathcal{C} \end{cases}$$

We think of this defining two *new* categories

1. The **reference** category $X = 0$.
2. The other category $X = 1$.

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Question: What if we had swapped the role of the reference category, so $X = 1$ when observation *not* in $\mathcal{C}$?

Answer: Not much. The $\beta_0 \rightarrow \beta_0 + \beta_1$ and $\beta_1 \rightarrow -\beta_1$.

Multiple Categories

What if we have a predictor which can belong to multiple categories?

The different categories are also called **levels**.

We proceed as you might expect. Say there are k levels.

1. Pick one level as the **reference** level.
2. Introduce $k - 1$ indicator variables for the remaining levels

The interpretation of each contrast is that it adjusts the intercept from the reference level to the coded level.

Example with Multiple Categories

We build a linear model for IQ of NPS students using a qualitative predictor of the student's curriculum.

- ▶ Y : IQ, a quantitative variable
- ▶ X : The student's curriculum; one of

{OR, Physics, Applied Math, Engineering, Other}

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- ▶ X_1 : 1 if student in OR, 0 if not.
- ▶ X_2 : 1 if student in Physics, 0 if not.
- ▶ X_3 : 1 if student in Applied Math, 0 if not.
- ▶ X_4 : 1 if student in Engineering, 0 if not.

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- ▶ X_4 : 1 if student in Engineering, 0 if not.

β_i is the **adjusted effect** of category i on β_0 .

Example with Multiple Categories

We can now predict IQ Y from categorical inputs.

$$Y = \beta_0 + X_1\beta_1 + X_2\beta_2 + X_3\beta_3 + X_4\beta_4$$

The fact that X_i are binary variables changes **nothing** about our linear regression developments, except that scatter plots against X should be replaced by box plots because X is discrete.

- ▶ For sample $(\mathbf{x}, \mathbf{y})$, form design matrix $\mathbf{x}$ and vector of coefficients $\hat{\beta}$.
- ▶ Still have $\hat{\beta} = (\mathbf{x}^T \mathbf{x})^{-1} \mathbf{x}^T \mathbf{y}$.
- ▶ In general, can have a **mix of categorical and numeric predictors**.

Linear Regression:

Predict quantitative variables using categorical¹ and quantitative variables.

¹By coding quantitative variables using binary predictors. 

An R Tutorial

R tricks

- ▶ Use `as.factor` to code a variable as factors (that is, levels of a qualitative variable).
- ▶ Use `relevel(x, ref=level1)` to set `level1` as reference category.

Next... an R tutorial.